

Takbir Zaman Bhuiyan

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PROFESSIONAL SUMMARY

Computer Science and Engineering graduate with hands-on industry experience as a Software Engineer, building scalable APIs and interactive web applications. Passionate about solving real-world problems through data-driven engineering while utilizing modern frameworks and AI-assisted development tools to deliver high-quality, full-stack solutions.

EDUCATION

2022 – 2025 **BSc in Computer Science & Engineering, AIUB** CGPA: 3.75/4.0

AWARDS

Dean's Award — Fall 2023–24, Fall 2024–25 (AIUB)

EXPERIENCE

Software Engineer Intern Feb 2025 – May 2025

NN Services & Engineering Ltd (NNSSEL) Dhaka, Bangladesh

- Built full pages and feature modules from scratch for the NNSSEL web platform using React (frontend) and Node.js/Express (backend), contributing directly to the production codebase.
- Designed and delivered assigned prototypes including a full-stack Todo application and a Login UI with authentication, which served as team-reviewed task deliverables.
- Participated in agile workflows, code reviews, and team collaboration across the full development lifecycle from design to deployment.

PROJECTS

RidePrice BD — ML Ride Fare Estimator Python, Flask, JavaScript [Live] [GitHub]

- Trained and deployed a Gradient Boosting model with feature engineering on real Dhaka ride data, delivering accurate fare predictions across car, CNG, bike, and bus.
- Architected a client-side JS fare engine alongside the Flask API, enabling instant predictions with zero server latency and full offline resilience.
- Shipped full-stack: Flask REST API on Render, Netlify-deployed frontend with demand-multiplier controls and a 76-location searchable dropdown.

Student Attendance Management C#, ASP.NET Core, SQL [GitHub]

- Architected a layered REST API using the repository pattern, EF Core, and strict API/BLL/DAL separation, with AutoMapper DTOs enforcing contract-safe data flow.

ConnectHub — Social Platform PHP, MySQL [Live] [GitHub]

- Built a modular PHP MVC platform across 9 controllers with a normalised MySQL schema, role-based access control, secure authentication, and session management.

Face Recognition System Python, OpenCV [GitHub]

- Constructed an OpenCV pipeline combining YuNet face detection and SFace embeddings, enhanced with CLAHE preprocessing, geometric alignment, and automated Colab execution.

beatMEE — 2D Browser Fighting Game HTML, CSS, JavaScript, PeerJS [Live] [GitHub]

- Built a browser-based 2D fighting game engine from scratch with frame-accurate combat mechanics, stamina systems, and adaptive AI using finite-state machine logic.
- Implemented P2P real-time multiplayer via WebRTC/PeerJS with multiple AI difficulty levels and a mobile virtual controller.

Hospital Management System Oracle [GitHub]

- Designed a normalised relational schema with referential integrity constraints, stored procedures, and optimised queries for multi-department hospital operations.

Online Job Management Portal Java, Swing [GitHub]

- Built a role-based Java Swing desktop application managing end-to-end job posting, application tracking, and hiring workflows across multiple user types.

RESEARCH EXPERIENCE

Explainable Osteoporosis Detection System DenseNet121, Grad-CAM

- Developed a DenseNet121 model for automated knee X-ray classification, achieving **87.5% recall** and enhancing clinical interpretability via Grad-CAM heatmap visualisation.

Waste Image Classification Using EfficientNet-B3 Deep Learning, Transfer Learning

- Designed a multi-class waste classifier using EfficientNet-B3 with transfer learning, achieving **94% accuracy** and outperforming baseline CNN models through optimised training and augmentation.

TECHNICAL SKILLS

Languages	C++, Python, C#, JavaScript, PHP, Java
Web & Frameworks	React, ASP.NET Core, Node.js, Express, Flask, HTML, CSS
Tools & DevOps	Git, GitHub, Vite, Netlify, Render, Google Colab
Databases	MySQL, Oracle, SQL Server (EF Core)
AI & Data Science	Deep Learning, Transfer Learning, Computer Vision, Predictive Modelling, Feature Engineering, Model Optimisation